

Customer Churn Prediction & LTV Engine

Week 3 • Day 3

FastAPI — Introduction & Project Setup

Build a REST API to serve your ML model predictions to any application

■ Day Objectives

- ◆ Understand REST APIs, HTTP methods, and JSON
- ◆ Install and configure FastAPI with Uvicorn
- ◆ Create the API project structure
- ◆ Build a health-check endpoint (/health)
- ◆ Understand Pydantic models for request/response validation

■ 1. What is a REST API?

An **API (Application Programming Interface)** is a way for two software systems to communicate. A **REST API** uses HTTP (the same protocol as websites) to exchange data. Your ML model becomes a service that any frontend, mobile app, or business system can call to get predictions.

HTTP Methods

- **GET:** Retrieve data. Like opening a webpage.
- **POST:** Send data to create/process something. Like submitting a form.
- **JSON:** JavaScript Object Notation — the standard data format for APIs. Looks like a Python dictionary.

FastAPI Overview

FastAPI is a modern Python web framework for building APIs. It automatically generates interactive documentation (Swagger UI) and uses type hints for validation. It is one of the fastest Python web frameworks available.

■■ 2. FastAPI Setup

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